P R a V A H

Predictive Routing and Vehicle's Augmented Handling

Intelligent Adaptive Traffic Management System for Enhanced Mobility

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Problem Statement

Urban traffic congestion, inefficient traffic signal timings, and delayed emergency vehicle response lead to increased travel times, accidents, pollution, and public safety risks. Current traffic points with hard-coded timers cannot keep pace with the ever growing complex traffic environments.

Solution Proposed

We propose— PRaVAH, an intelligent, sensor-driven traffic management system that dynamically recognizes and analyzes road infrastructure, traffic flow, and environmental conditions to optimize traffic signal timings and vehicle speed limit to improve overall traffic efficiency and public safety. Key features include real-time analysis of vehicle and pedestrian flows, emergency vehicle detection, disruption prediction (accidents, rallies, fights), and adaptation to special cases such as political convoys and festivals. By integrating weather data and detailed map information, the system adjusts signal durations dynamically, ensuring safer and smoother traffic movement. Notifications of disruptions are provided to relevant authorities for swift response, and data is continuously collected for predictive analytics and long-term improvements.

Implementation Plan

The Vision

- The system uses a combination of optical cameras for object tracking and radar sensors for distance and speed measurement under different visibility conditions

- The system will initially be deployed at traffic signals (in urban or semi-urban cities) prone to congestion and accidents
- Beneficiaries include daily commuters, emergency services, pedestrians and traffic and criminal law enforcement institutions.

The project will proceed through phases:

1. SITL:
    a. Data acquisition for simulation
    b. Training ML model for traffic analysis and signal optimization
    c. Testing in simulation environment

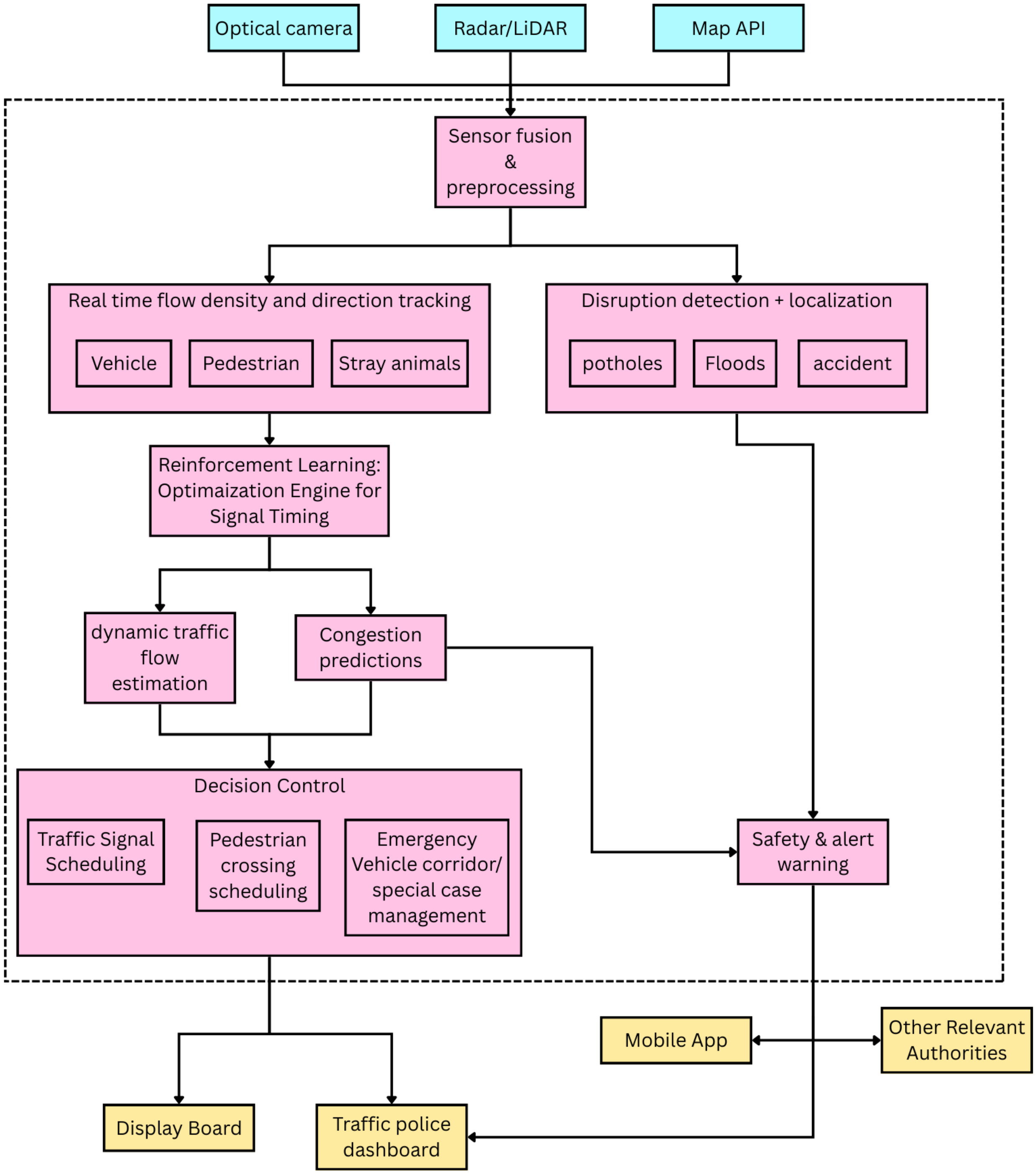
2. HITL:
    a. Sensor installation and integration
    b. Real data collection
    c. Iterative refinement based on real-world feedback

Deliverables include:

1. Predictive Traffic management device placed at the traffic signal (along with a large display for the commuters):
    a. Optimized traffic signal timings
    b. Additional pedestrian crossing timings
    c. Alerts for presence of nearby potholes, floodings, animals, accidents, etc.
2. Dashboard for traffic police to monitor real-time and notifications.

Expected outcomes and Potential real - impact

- reduced congestion
- improved emergency response times
- decreased accident rates
- enhanced commuter satisfaction
- Tracking criminal vehicles by combining data from a network of PRaVAH devices installed at multiple locations.
- AI-driven predictive analytics for short-term traffic trend forecasting, such as- office hours, weekends, etc.


Optical camera
Radar/LiDAR
Map API
Sensor fusion
&
preprocessing
Real time flow density and direction tracking
Vehicle
Pedestrian
Stray animals
Disruption detection + localization
potholes
Floods
accident
Reinforcement Learning:
Optimaization Engine for
Signal Timing
dynamic traffic
flow
estimation
Congestion
predictions
Decision Control
Traffic Signal
Scheduling
Pedestrian
crossing
scheduling
Emergency
Vehicle corridor/
special case
management
Safety & alert
warning
Mobile App
Other Relevant
Authorities
Display Board
Traffic police
dashboard

Safety and Efficiency Analysis

- Efficiency is improved by drastically reducing unnecessary wait times at traffic signals by dynamically adjusting signal durations
- The solution enhances safety by enabling prioritized movement for emergency vehicles and adapting traffic controls to reduce congestion-related incidents.
- Continuous monitoring of pedestrian flow and vehicle types allows safer crossing times lane usage.
- Automated alerts for disruptions facilitate faster incident management, minimizing traffic jams and associated risks.
- Robust to difficult weather conditions, such as- fog, rain, etc.
- Uses multiple sensors to ensure safe and precise functioning during sensor failure.

Scalability and Future Development

The system architecture is designed for scalability, allowing expansion to multiple intersections and integration with city-wide traffic control centers. Future enhancements include incorporation of AI-driven predictive analytics for long-term traffic trend forecasting, integration with vehicle-to-infrastructure (V2I) communication systems for smarter vehicle coordination, and the inclusion of additional sensor types such as LIDAR and audio sensor for more precise environment mapping and analysis. The platform can be adapted for use in diverse environments, including highways, smart cities, rural intersections, and event-specific traffic management scenarios.

Innovation Advantage

- Holistic Real-Time SMART Traffic Solution for managing and optimizing traffic flow.
- Multiple Sensor Fusion for enhanced and robust performance in different scenarios and weather conditions.
- Real-Time Adaptive Response which not only detects problems but actively mitigates them through automatic rerouting and IoT-enabled signage.
- AI-Powered Road Hazard Recognition capable of detecting non-standard obstacles common in India, such as cattle, street vendors, and potholes, not just vehicles.
- Low-Cost Deployment by Utilizing a hybrid mesh of affordable sensors and leverage resources such as Google Maps making it scalable to Tier-2/Tier-3 cities.
- Uses machine learning to predict potential issues like accidents, waterlogging, or congestion, enabling pre-emptive alerts to authorities.
- Can easily integrate with existing traffic infrastructure (traffic lights, toll booths) without requiring a full revamp.

- Automatically detects emergency vehicles and makes way for them on priority, in real-time.
- Addresses the needs of a wide range of users, including vehicles (cars, trucks, two-wheelers), pedestrians, and law enforcement.
- Possesses real-time environmental and temporal awareness to make informed decisions.